GOVERNMENT OF MAHARASHTRA

MAHA-GR-ALIGN

AI-Assisted GR Drafting & Alignment

Technical Walkthrough · Architecture, Features, Stack & Live Demo

6-MINUTE PRESENTATION + LIVE DEMO

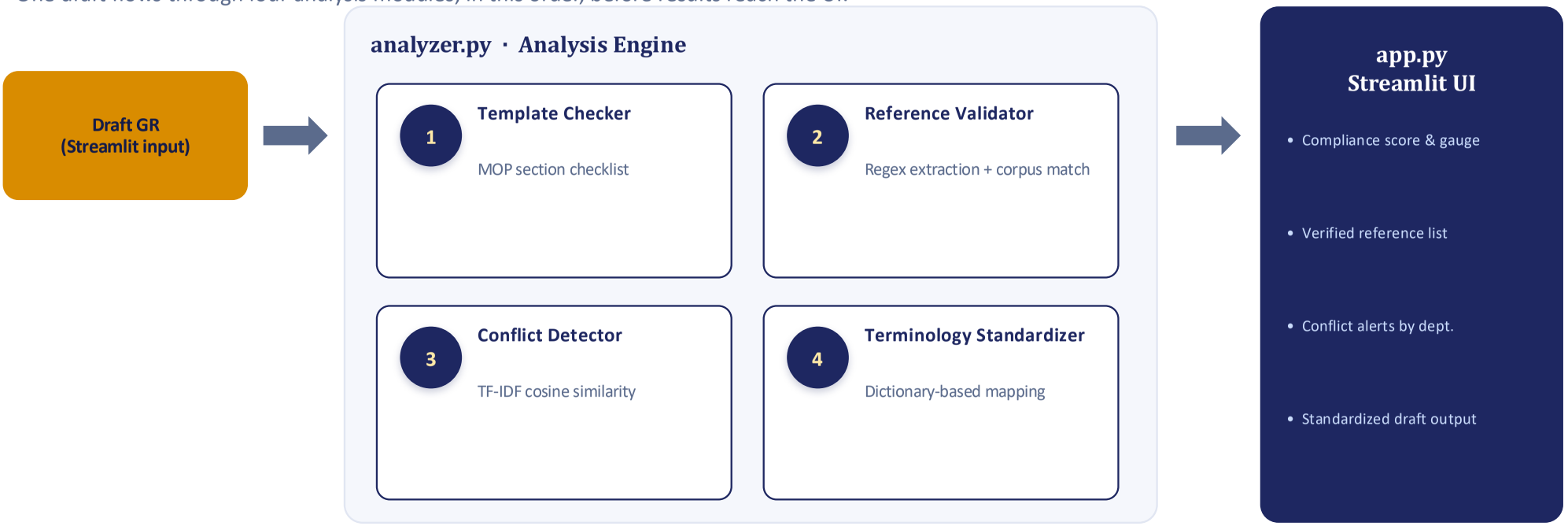
TEAM BLITZKRIEG

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PROBLEM AREA 1

System Architecture & Workflow

One draft flows through four analysis modules, in this order, before results reach the UI.



Request → Response Workflow

Draft submitted → 4 modules run → JSON result built → UI renders instantly

Key Features & Alignment with Requirements

Every Problem Area 1 requirement maps to a specific, working module — not a placeholder.

FEATURE	HOW IT WORKS	REQUIREMENT SATISFIED
Cross-Departmental Conflict Detection	TF-IDF vectorization + cosine similarity against the GR corpus, plus rule-based contradiction heuristics (e.g. “without concurrence” vs. “concurrence required”).	Cross-Departmental Conflict Detection
Automated Reference Tracking	Regex extraction of every cited GR number and date, cross-checked against the live corpus for a verified / unverified status.	Automated Reference Tracking
Strict Template Enforcement	Each draft is scored section-by-section against a configurable MOP checklist (PREAMBLE, REFERENCE, DECISION, SANCTION, ORDER, DISTRIBUTION).	Strict Template Enforcement (MOP)
Terminology Standardization	Dictionary-backed mapping flags informal phrasing and auto-generates a standardized rewrite of the draft.	Standardized Legal Terminology

Technology Stack: Open-Source & APIs

A fully open-source, auditable stack — with zero proprietary or paid API dependency.

✓ Open-Source Stack (all in use today)

Python 3
Core application language

Streamlit
Interactive web front-end

scikit-learn
TF-IDF vectorization & cosine similarity

Regex (re)
GR-number & date extraction

Plotly
Interactive compliance gauge

JSON
Corpus, MOP template & term dictionary

✗ Proprietary APIs / Paid Services

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NONE USED

No external API calls, no cloud LLM, no per-request billing, and no vendor lock-in — every module runs on standard, auditable open-source libraries.

Local / Offline Deployment Capability

Built to run entirely on-premise — a requirement, not an afterthought, for government resolution data.

No Internet Required at Runtime

Every module — template check, reference match, TF-IDF conflict scoring, terminology lookup — runs on local Python and local JSON files.

Data Never Leaves the Machine

The GR corpus, MOP rules, and terminology dictionary are plain local files — suitable for an air-gapped State Data Centre / NIC deployment.

Single-Command Local Launch

streamlit run app.py — no container orchestration or cloud account needed to stand up a working instance today.

Ready for a Local LLM Upgrade

The architecture already isolates each check into a swappable module — a future semantic layer (e.g. via Ollama running a local model) can slot in without ever calling out to the internet.

Live Demonstration

Switching to the running app now — here's exactly what we'll show, in order.

1	**Fully Compliant Draft**	Load the scenario → 100% MOP compliance, both references verified, zero conflicts, clean terminology.
2	**Cross-Department Conflict Draft**	Load the scenario → engine flags a real contradiction with a Finance Department concurrence rule.
3	**Missing-Sections Draft**	Load the scenario → compliance gauge drops into the red; missing sections listed explicitly.
4	**Light / Dark Theme + Sidebar**	Toggle appearance and collapse/expand the sidebar to show the interface is fully interactive.



Thank You

Questions?

MAHA-GR-ALIGN · Team Blitzkrieg · VJTI AI Hackathon 2026